

Evaluation report

Physics · quantitative · grade band 11-12

Verdict

Stage score	95.9 / 100 (exemplary)
Confidence	92%
Stages scored	10 of 11
Rubric score	93.3 / 100 (exemplary)
Package	b5138b0c-295d-405e-bf4a-8348d0846724
Evaluated	2026-08-04T04:38:07+00:00

Two scores, not one. The stage score asks whether each pipeline stage met its contract; the rubric score asks whether the result is good teaching. They are reported separately because averaging them would produce a number that answers neither question.

Against history

1 comparable run(s) on record; a baseline needs at least 5 before a fall in score can be told apart from run-to-run variation.

Stages

Stage	Score	Conf.	Measured	Unmeasurable
Document intelligence	95.2	50%	3	3
Educational classification	98.5	80%	4	1
Knowledge extraction	86.5	100%	5	0
Teaching planner	100.0	100%	3	0
Lesson generation	100.0	100%	4	0
Activity generation	100.0	100%	6	0
Assessment generation	100.0	86%	6	1
Gap analysis	95.1	100%	5	0
Validation	87.0	100%	4	0
Publishing	97.3	100%	5	0
Learning outcomes	n/a	0%	0	4

Not measurable (9)

These metrics carry no score. Each one states what it would take to measure — a labelled corpus, a classroom trial, response data. Reporting a number here would mean inventing one.

OCR confidence (document-intelligence)

No page needed OCR: every page carried a text layer, so nothing was read from an image and there is no confidence to report. Measuring it would need: a document with at least one scanned page

OCR accuracy (document-intelligence)

Accuracy is a comparison against what the page actually said, and no uploader supplies a transcript of their own scan. Confidence is reported instead, which is the recogniser's certainty, not its correctness — the two come apart exactly when it matters. Measuring it would need: reference transcriptions of the scanned pages to diff against

Extraction fidelity vs. original (document-intelligence)

Fidelity is a comparison, and the original document's ground-truth text is not part of the package. Measuring it would need: a hand-checked transcript of the source PDF to diff the extraction against

Subject / grade classification accuracy (educational-classification)

Accuracy needs labels. There is no corpus of documents with agreed subject and grade-band annotations to score this run against. Measuring it would need: a held-out set of documents labelled by teachers, ideally 100+ across subjects

Item difficulty is calibrated (assessment-generation)

Difficulty is a property of a population's responses. No student has answered these items, so there is no p-value or discrimination index to report. Measuring it would need: response data from a cohort — item-response theory needs roughly 100 students per item

Teacher satisfaction (outcomes)

Satisfaction is a rating teachers give. None have used this package. Measuring it would need: a pilot with teachers rating packages after classroom use — 20+ responses before a mean means anything

Student learning effectiveness (outcomes)

Effectiveness is measured by what students learn, which requires students, a pre-test and a post-test. Measuring it would need: a controlled classroom trial with pre/post assessment against a conventionally-planned unit

Teacher preparation time saved (outcomes)

Time saved is a difference against how long the same teacher would take unaided. That baseline has not been collected. Measuring it would need: timed preparation of the same unit by teachers with and without the package

Conformance to a published curriculum (outcomes)

The package aligns to a board's structure, but no authority has certified the mapping. Reporting a conformance score would state agreement nobody granted. Measuring it would need: an official syllabus document parsed to statement level, and a reviewer's sign-off on the mapping

Recommendations (4)

High

Validation — No blocking issues remain (57.1)

Route the blocking issues back to their owning stage for repair.

Why it matters: The package ships with defects its own validator found.

Knowledge extraction — Quotes appear in the chunks they cite (59.5)

Re-run extraction with the citation verifier rejecting spans whose quote is not substring-present in the cited chunk.

Why it matters: A citation that does not check out is worse than no citation — a teacher trusts it.

Medium

Publishing — Every stage reported its cost (90.0)

Emit a timing entry from every stage span, including cheap ones.

Why it matters: Per-stage cost cannot be attributed without it.

Low

Document intelligence — Source metadata captured (85.7)

Record detected_language at parse time.

Why it matters: Provenance cannot be audited without it.

Metric detail

Document intelligence — 95.2

Source metadata captured — 85.7 (measured, confidence 1.00)

6 of 7 source fields recorded; missing detected_language.

- /source — recorded: filename, mime, sha256, size_bytes, page_count, word_count

Text extracted per page — 100.0 (measured, confidence 1.00)

468 words per page across 44 pages. Consistent with a born-digital document.

- /source/word_count — 20,603 words over 44 pages

Chunks carry text — 100.0 (measured, confidence 1.00)

63 of 63 chunks hold text. Empty chunks break every citation that lands on them.

- /chunks — 63 chunks retained

OCR confidence — n/a (not_measurable, confidence 0.00)

No page needed OCR: every page carried a text layer, so nothing was read from an image and there is no confidence to report. Measuring it would need: a document with at least one scanned page

OCR accuracy — n/a (not_measurable, confidence 0.00)

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Extraction fidelity vs. original — n/a (not_measurable, confidence 0.00)

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Educational classification — 98.5

Classification fields populated — 100.0 (measured, confidence 1.00)

7 of 7 fields classified.

- /classification — subject='Physics', profile='quantitative'

Pedagogy profile is one the pipeline routes on — 100.0 (measured, confidence 1.00)

Profile 'quantitative' is a routable profile; every downstream stage branches on it.

- /classification/pedagogy_profile — quantitative

Classifier's own confidence — 83.0 (measured, confidence 1.00)

Mean self-reported confidence 0.83 across category, chapter, curriculum_alignment, difficulty, grade_band, language, pedagogy_profile, subject, topic. This is the classifier's own estimate, not an accuracy measurement.

- /classification/confidences — {'subject': 0.95, 'grade_band': 0.7, 'difficulty': 0.8, 'topic': 0.95, 'chapter': 0.9, 'category': 0.9, 'language': 0.99, 'pedagogy_profile': 0.98, 'curriculum_alignment': 0.3}

Low-confidence fields flagged for review — 100.0 (measured, confidence 1.00)

Every field scored below 0.7 is flagged for review.

- /classification/low_confidence_fields — ['curriculum_alignment']

Subject / grade classification accuracy — n/a (not_measurable, confidence 0.00)

Accuracy needs labels. There is no corpus of documents with agreed subject and grade-band annotations to score this run against. Measuring it would need: a held-out set of documents labelled by teachers, ideally 100+ across subjects

Knowledge extraction — 86.5

Concepts carry evidence — 100.0 (measured, confidence 1.00)

22 of 22 concepts carry evidence.

Concepts carry a summary — 100.0 (measured, confidence 1.00)

22 of 22 concepts carry summary.

Quotes appear in the chunks they cite — 59.5 (measured, confidence 1.00)

22 of 37 evidence spans quote text that is present verbatim in the chunk they cite. 15 do not.

- /knowledge/concepts/2/evidence — quote not found in c_0003: 'It is not possible to create or destroy net charge carried b'
- /knowledge/concepts/3/evidence — quote not found in c_0003: 'Experimentally it is established that all free charges are i'
- /knowledge/concepts/5/evidence — quote not found in c_0009: 'Coulomb's force law between two point charges q and q locate'
- /knowledge/concepts/5/evidence — quote not found in c_0009: 'Equation (1.3) is valid for any sign of q and q whether posi'
- /knowledge/concepts/6/evidence — quote not found in c_0011: 'Experimentally, it is verified that force on any charge due\n'

Objectives carry a Bloom level — 100.0 (measured, confidence 1.00)

7 of 7 objectives are tagged with a Bloom level; 4 distinct level(s) used.

- /knowledge/learning_objectives — levels: ['analyze', 'apply', 'remember', 'understand']

Concept graph resolves to real concepts — 100.0 (measured, confidence 1.00)

8 edges over 22 nodes. Every endpoint resolves to an extracted concept.

Teaching planner — 100.0

Every concept is scheduled into a period — 100.0 (measured, confidence 1.00)

22 of 22 concepts are taught in some period.

- /teaching_plan/periods — 6 periods scheduled

Prerequisites are taught before what needs them — 100.0 (measured, confidence 1.00)

8 of 8 prerequisite edges are respected by the period order.

Period plans fit the period — 100.0 (measured, confidence 1.00)

6 of 6 periods allocate no more than the 40-minute period.

Lesson generation — 100.0

Every planned period has classroom content — 100.0 (measured, confidence 1.00)

6 of 6 planned periods have generated content.

- /classroom_content — 6 content blocks for 6 periods

Each period carries the full set of lesson blocks — 100.0 (measured, confidence 1.00)

36 of 36 lesson blocks are populated across 6 periods.

Teacher scripts have enough to work from — 100.0 (measured, confidence 1.00)

Teacher scripts average 1343 words per period against a 250-word working floor. This measures volume, not teaching quality.

- /classroom_content/0/teacher_script — 1123 words

Activity references resolve — 100.0 (measured, confidence 1.00)

6 of 6 activity references point at an activity that exists.

Activity generation — 100.0

Activities vary in kind — 100.0 (measured, confidence 1.00)

5 distinct activity type(s) across 6 activities (demonstration, experiment, problem_set, simulation, think_pair_share), against a target of 4.

- /activities — types: ['demonstration', 'experiment', 'problem_set', 'simulation', 'think_pair_share']

Activities target extracted concepts — 100.0 (measured, confidence 1.00)

6 of 6 activities reference at least one concept that stage 3 actually extracted.

Activities carry teacher instructions — 100.0 (measured, confidence 1.00)

6 of 6 activities carry teacher_instructions.

Activities state success criteria — 100.0 (measured, confidence 1.00)

6 of 6 activities carry success_criteria.

Activities carry differentiation — 100.0 (measured, confidence 1.00)

6 of 6 activities carry differentiation.

Activities fit inside a period — 100.0 (measured, confidence 1.00)

6 of 6 activities run within the 40-minute period.

Assessment generation — 100.0

Every item can be marked — 100.0 (measured, confidence 1.00)

24 of 24 items carry an answer or a rubric.

MCQs have one right answer and real distractors — 100.0 (measured, confidence 1.00)

7 of 7 multiple-choice items have three or more options with exactly one marked correct.

Objectives are assessed — 100.0 (measured, confidence 1.00)

7 of 7 learning objectives are reachable from at least one assessment item.

Items span more than recall — 100.0 (measured, confidence 1.00)

4 distinct Bloom level(s) in the bank (analyze, apply, remember, understand). Three or more is the point at which a paper distinguishes recall from reasoning.

- /assessments/items — bloom levels: ['analyze', 'apply', 'remember', 'understand']

Blueprint matches the items actually present — 100.0 (measured, confidence 1.00)

The published blueprint agrees with a recount of the bank.

- /assessments/blueprint/items_by_kind — {'numerical': 10, 'mcq': 7, 'short_answer': 5, 'long_answer': 2}

Total marks equal the sum of the items — 100.0 (measured, confidence 1.00)

Published total is 74; the items sum to 74. They agree.

- /assessments/total_marks — 74 vs 74 summed

Item difficulty is calibrated — n/a (not measurable, confidence 0.00)

Difficulty is a property of a population's responses. No student has answered these items, so there is no p-value or discrimination index to report. Measuring it would need: response data from a cohort — item-response theory needs roughly 100 students per item

Gap analysis — 95.1

Gaps attach to real concepts — 100.0 (measured, confidence 1.00)

6 of 6 predicted gaps name a concept that stage 3 extracted.

Gaps carry a remediation — 100.0 (measured, confidence 1.00)

6 of 6 gaps carry remediation.

Gaps carry a diagnostic question — 100.0 (measured, confidence 1.00)

6 of 6 gaps carry diagnostic_questions.

Diagnostics predict the wrong answer — 100.0 (measured, confidence 1.00)

12 of 12 diagnostic questions state the wrong answer the misconception produces, which is what makes them diagnostic rather than merely difficult.

- /learning_gaps — 12 diagnostic questions

Gaps trace to extracted misconceptions — 16.7 (measured, confidence 1.00)

1 of 6 gaps restate a misconception stage 3 found in the source. The remainder are predicted rather than sourced, which is a legitimate output of this stage.

- /knowledge/misconceptions — 2 extracted misconceptions

Validation — 87.0

No blocking issues remain — 57.1 (measured, confidence 1.00)

Status 'fail' with 7 issue(s), 3 of them blocking.

- /validation/issues — GROUNDING_UNSUPPORTED_CLAIM: claim at /knowledge/concepts/13 is not supported by its cited source (judge rated this unsupported)
- /validation/issues — GROUNDING_UNSUPPORTED_CLAIM: claim at /knowledge/formulae/10 is not supported by its cited source (judge rated this unsupported)
- /validation/issues — GROUNDING_UNSUPPORTED_CLAIM: claim at /knowledge/applications/0 is not supported by its cited source (judge rated this

unsupported)

Coverage report survives a recount — 100.0 (measured, confidence 1.00)

The published coverage numbers match an independent recount of the package.

- /validation/coverage — 22/22 concepts taught, confirmed

Structural checks pass — 100.0 (measured, confidence 1.00)

3 of 3 structural checks are clean.

Claims are supported by the source — 91.1 (measured, confidence 1.00)

Stage 9 scored grounding at 0.91. 7 claim(s) were recorded as unsupported.

- /validation/unsupported_claims — Gauss's law: Electric flux through a closed surface equals the total charge enclosed divided by ϵ_0 .
- /validation/unsupported_claims — Torque on a dipole in a uniform field: A permanent dipole p in a uniform electric field E experiences a torque $\tau = p \times E$
- /validation/unsupported_claims — Continuous charge distributions (linear, surface, volume): Macroscopic charge distributions are described by linear (λ),
- /validation/unsupported_claims — Dipole far-field (axis) E (on axis, $r \gg a$) = $(1/4 \pi \epsilon_0) (2 p / r^3)$ along p
- /validation/unsupported_claims — Dipole far-field (equatorial) E (on equatorial plane, $r \gg a$) = $-(1/4 \pi \epsilon_0) (p / r^3)$

Publishing — 97.3

Every stage reported its cost — 90.0 (measured, confidence 1.00)

9 of 10 stages recorded a timing entry; missing publishing.

- /provenance/stage_timings — 9 entries totalling 549444 ms

Each model-calling stage names the model that ran it — 100.0 (measured, confidence 1.00)

8 of 8 model-calling stages record which model produced their output. document-intelligence, publishing are excluded because they make no model call, so an absent entry for them is correct rather than missing.

- /generator/models_by_stage — {'educational-classification': 'gpt-5-mini-2025-08-07', 'knowledge-extraction': 'gpt-5-mini-2025-08-07', 'teaching-planner': 'gpt-5-mini-2025-08-07', 'lesson-generation': 'gpt-5-mini-2025-08-07', 'act

Stages explain their choices — 100.0 (measured, confidence 1.00)

7 stage decision(s) recorded with a reason. Four is the point at which the main routing choices — period count, item mix, gap severity, profile — are all accounted for.

- /provenance/decisions — knowledge-extraction: extraction split into 3 passes — 28,244 tokens exceeds the 12,000-token window this route leaves for document text after the schema, system prompt, and the reservation for the answer; sections are packed to that window rather than one call each, so cost follows content volume
- /provenance/decisions — knowledge-extraction: pedagogy pass reads 16 of 63 chunks — objectives and edges are document-level judgements, but the document is 28,244 tokens against a 12,000-token window; scope comes from the full 26-concept inventory, which is sent whole, and the text is narrowed to the passages those concepts cite plus a stride across the rest, so every claim still has a quotable source
- /provenance/decisions — knowledge-extraction: 5 claims dropped — their quotes did not

appear in the chunk they cited; an unverifiable citation is removed here rather than propagating into six later stages

- /provenance/decisions — teaching-planner: 6 periods of 40 minutes — derived from 22 concepts weighted by importance against a 40-minute period

Package is identified and versioned — 100.0 (measured, confidence 1.00)

3 of 3 identity fields present (schema 1.0.0).

- /schema_version — 1.0.0

Cost and token usage are accounted for — 100.0 (measured, confidence 1.00)

\$0.0000 across 165,079 input and 107,367 output tokens.

- /provenance/total_cost_usd — \$0.0000

Learning outcomes — n/a

Teacher satisfaction — n/a (not_measurable, confidence 0.00)

Satisfaction is a rating teachers give. None have used this package. Measuring it would need: a pilot with teachers rating packages after classroom use — 20+ responses before a mean means anything

Student learning effectiveness — n/a (not_measurable, confidence 0.00)

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